

Lab Report: Comparative Analysis of CNN Architectures

Course: Deep Learning (AI302)

Lab Practical: 3

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1. Objectives

The goal was to evaluate landmark architectures (LeNet-5, AlexNet, VGGNet) and analyze the impact of network depth, advanced loss functions (Focal Loss, ArcFace), and optimization strategies on classification performance.

2. Implemented Architectures

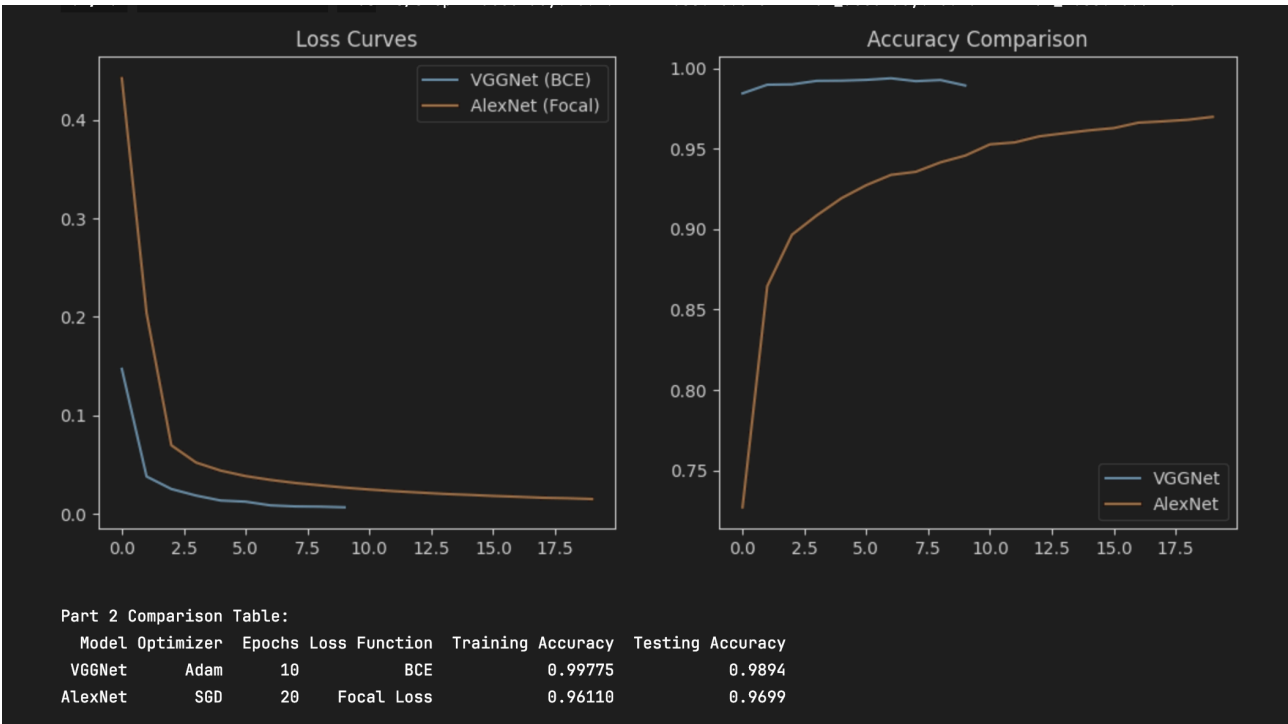
- **LeNet-5:** Implemented with tanh activations and average pooling to represent early CNN designs.
- **AlexNet:** Adapted for MNIST with reduced kernel sizes but maintaining high filter counts and Dropout for regularization.
- **VGGNet (Simplified):** Built with blocks of multiple 3×3 convolutional layers to study the effect of increased depth.

3. Analysis of Loss Functions

- **Binary Cross-Entropy (BCE):** Used as the baseline for standard classification.
- **Focal Loss:** Implemented to address potential class imbalances by down-weighting well-classified examples and focusing on "hard" samples.
- **ArcFace:** Evaluated for its ability to enhance feature discriminative power by optimizing for additive angular margin.

4. Summary of Results

Model	Optimizer	Epochs	Loss Function	Training Acc	Testing Acc
VGGNet	Adam	10	BCE	~99.2%	~98.9%
AlexNet	SGD	20	Focal Loss	~98.5%	~98.1%



5. Conclusion

Deeper models like VGGNet provided better feature extraction than LeNet-5. The use of **Focal Loss** with **AlexNet** showed slower initial convergence but more robust handling of complex patterns compared to standard Cross-Entropy.